ABDULRAHMAN ALSAADI

MECHATRONICS ENGINEER | ROBOTICS & AUTONOMOUS SYSTEMS

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PROFESSIONAL SUMMARY

CSWP-certified Mechatronics Engineering graduate with hands-on experience developing robotic systems from mechanical design and simulation to embedded integration and physical testing. Contributed across the development of a TEKNOFEST autonomous UAV, including mission software, onboard hardware integration, computer vision, and five flight-test sessions. Independently designed and built a 3-DOF SCARA robotic arm with a verified 492.5 mm reach and reduced one arm link's mass by 16.5% through topology optimization.

TECHNICAL SKILLS

Robotics & Autonomy: ROS 2, Gazebo, RViz, URDF/Xacro, ArduPilot SITL, MAVLink, pymavlink, Forward & Inverse Kinematics
Embedded Systems & Control: ESP32, Raspberry Pi, Arduino, Cube Orange, UART, MQTT, Stepper/Servo/BLDC Motors
Programming & Computer Vision: C++, Python, OpenCV, Linux (Ubuntu)
Mechanical Design & Prototyping: SolidWorks, SolidWorks Simulation (FEA), Topology Optimization, DFM, Dimensional Tolerancing, Bearing Fits, BOM Development, 3D Printing, Manufacturing Drawings

SELECTED ENGINEERING PROJECTS

WD DRONE - TEKNOFEST 2026 UAV Competition View Project — Karabük, Türkiye

UAV & Robotics Engineer — **Feb 2026 - Jul 2026**

- Developed and validated autonomous mission functions in Gazebo and ArduPilot SITL, covering waypoint search, visual target detection and approach, payload release, and return-to-launch; transferred mission logic to the Raspberry Pi 5 and Cube Orange platform.
- Integrated Raspberry Pi 5, Camera Module 3, and Cube Orange via MAVLink over UART using pymavlink; implemented OpenCV-based target detection at approximately 30 FPS, flight-mode switching, and payload-release control.
- Supported five UAV flight-test sessions, including a 12-minute unloaded hover and a 7-minute hover carrying a 1 kg payload; investigated flight oscillations through PID tuning.
- Contributed to end-to-end UAV development, from system requirements, component selection, BOM development and procurement to hardware integration and flight testing, while coordinating engineering tasks within a six-member team.

SCARA ROBOTIC ARM - GRADUATION PROJECT View Project — Karabük, Türkiye

Mechatronics & Robotics Engineer — **Apr 2026 - Jun 2026**

- Independently designed, manufactured, and validated a custom 3-DOF SCARA robotic arm from concept to physical prototype, achieving a verified 492.5 mm reach and 100 g payload at full extension.
- Reduced Arm Link 1 mass by approximately 16.5% using SolidWorks FEA and topology optimization; designed a two-stage GT2 belt transmission, calculated joint torques, and produced functional PLA+ components and manufacturing drawings.
- Developed ESP32/C++ control with automatic homing, forward and inverse kinematics, Cartesian positioning, and servo-gripper actuation; integrated the arm via MQTT with a vision-enabled conveyor and mobile robot for automated box transfer.

EXPERIENCE

SMART METHODS View Work — Jeddah, Saudi Arabia

Robotics Engineering Intern — **Jul 2025 - Aug 2025**

- Developed a web interface and Flutter app for robot-arm control, enabling the operator to save, replay, and manage movements using MySQL and HTTP over local Wi-Fi.
- Built Python and Flutter-based AI assistant prototypes integrating the Cohere API, speech-to-text, and text-to-speech for voice and text interaction; tested and debugged Android application functionality.
- Designed a 5-DOF robotic-arm concept in SolidWorks and separately assembled a 3D-printed 4-DOF robotic arm prototype with servo actuation and ESP32/Arduino electronics.

WASH THRU View Work — Jeddah, Saudi Arabia

Industrial Automation Panel Assembly Intern — **Jun 2025 - Jul 2025**

- Assembled and wired industrial control panels from electrical schematics, working with a Siemens PLC, I/O expansion modules, VFDs, contactors, relays, and motor-protection devices.
- Prepared, crimped, and terminated wires; inspected wiring against schematics, supported corrective actions, and documented assembly activities for technical reporting.

EDUCATION

Bachelor of Science (B.Sc.) in Mechatronics Engineering — **Jun 2021 - Jun 2026**
Karabük University, Türkiye

CERTIFICATIONS & PROFESSIONAL MEMBERSHIPS

- SOLIDWORKS: Simulation for Finite Element Analysis Course, LinkedIn Learning | Oct 2026
- Engineer Membership (Active), Saudi Council of Engineers (SCE) | Sep 2026
- Certified SOLIDWORKS Professional - Mechanical Design (**CSWP**), Dassault Systèmes | Aug 2026
- Certified SOLIDWORKS Associate - Mechanical Design (**CSWA**), Dassault Systèmes | May 2026

LANGUAGES

Arabic: Native | English: Upper-intermediate | Turkish: Intermediate